



Vantage® Research Ultrasound System

Vantage System Troubleshooting Guide

D00022 Rev. A

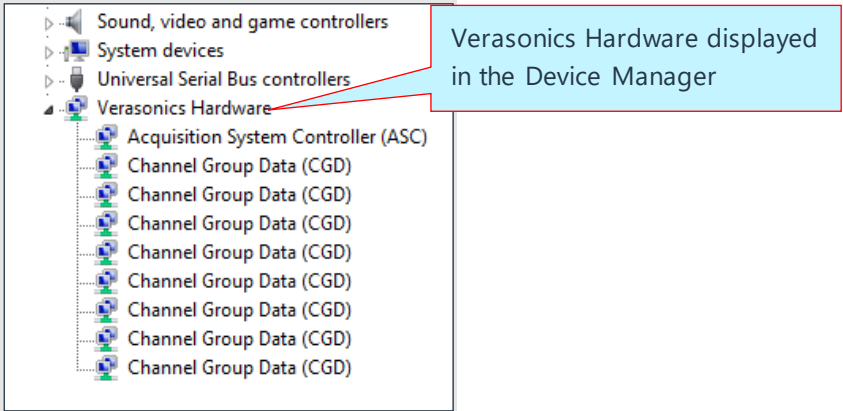
Summary

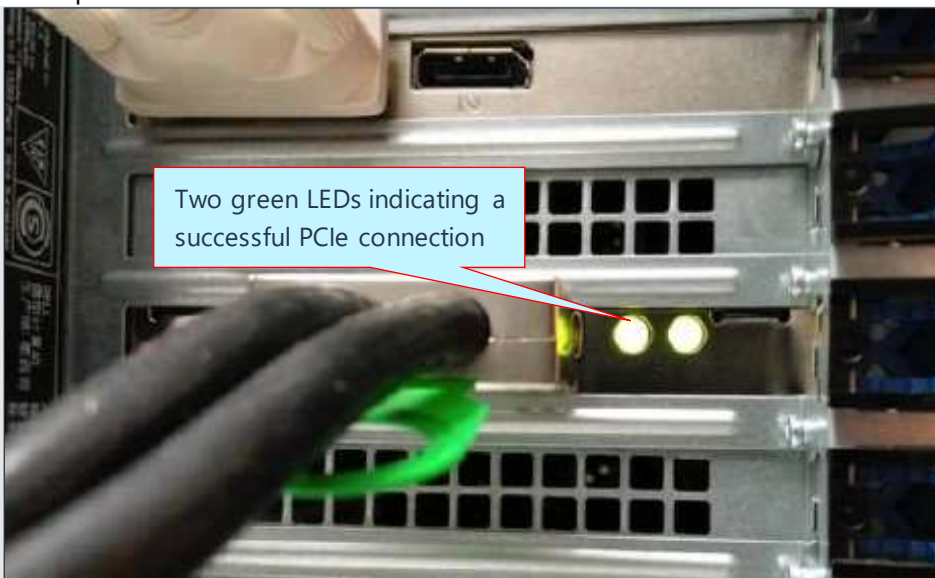
This document outlines troubleshooting procedures for some common hardware and software issues. In addition, the document provides instructions on how to collect the troubleshooting logs and test results before sending these files as an email attachment to the [Verasonics Support team](#).

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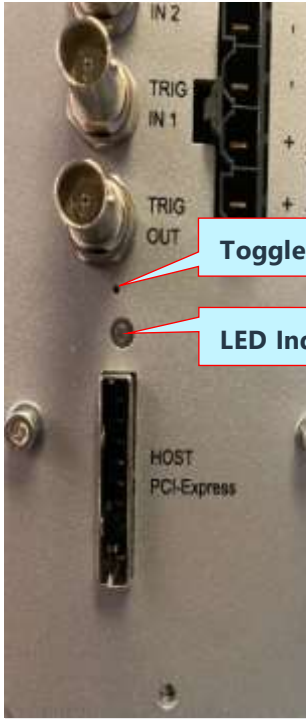
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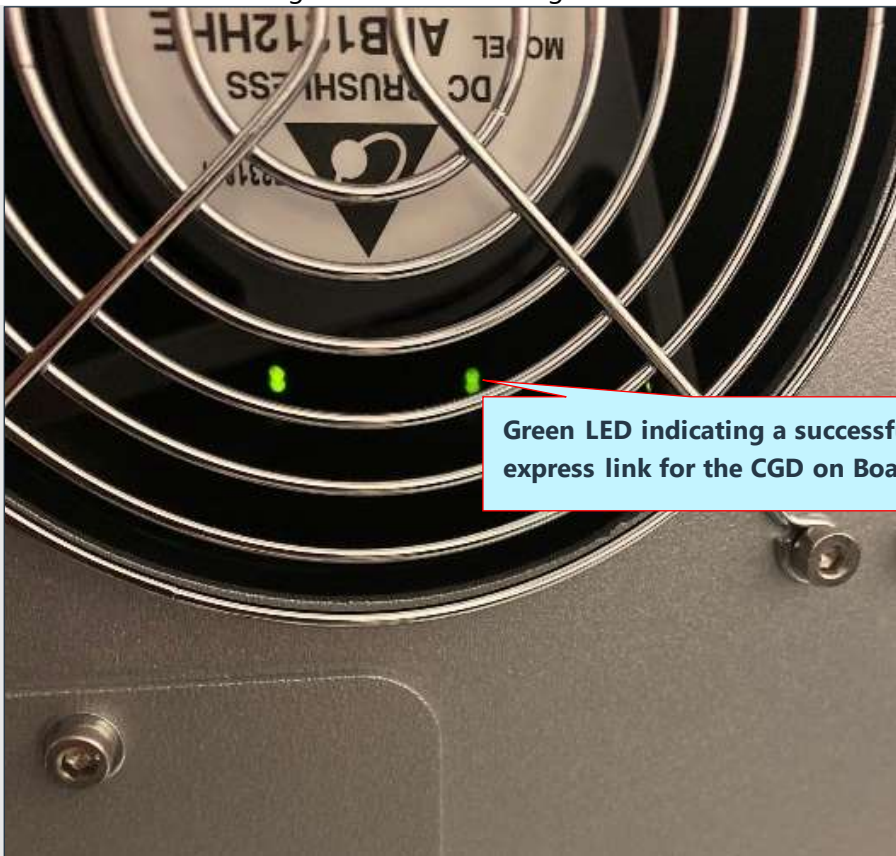
1. Troubleshoot Hardware Issues

Step#	Issue	Troubleshooting Procedure															
1	When MATLAB crashes during runtime and the Vantage system becomes unavailable with an error message, "The hardware can be detected but not initialized."	Reset the system by performing a cold boot: 1. Power off the host controller. Note: Ensure that the host controller is completely powered off and not restarted—restating the host controller will not fix the issue. 2. Power off the Vantage hardware system. 3. Wait for a few seconds and turn the power back on the host controller and Vantage hardware system. If the fault is not resolved after performing a cold boot, proceed to Step 2.															
2	When MATLAB crashes during runtime and the Vantage system becomes unavailable with an error message, "The hardware can be detected but not initialized."	Open Device Manager and ensure that the Verasonics Hardware is listed in the Device Manager. 1. Open the Device Manager. 2. Note that the Device Manager opens displaying a list of hardware devices.  Figure 1: Verasonics Hardware as displayed in the Device Manager 3. Count the number of Channel Group Data (CGDs) and Acquisition System Controllers (ASCs) listed under the Verasonics Hardware entry. 4. Compare the number of CGDs and ASCs expected in your Vantage System configuration. Use the table below. <table border="1"> <thead> <tr> <th>Vantage System Configuration</th><th>ASCs</th><th>CGDs</th></tr> </thead> <tbody> <tr> <td>Vantage 64</td><td>1</td><td>2</td></tr> <tr> <td>Vantage 64LE</td><td>1</td><td>4</td></tr> <tr> <td>Vantage 128</td><td>1</td><td>4</td></tr> <tr> <td>Vantage 256</td><td>1</td><td>8</td></tr> </tbody> </table> Table 1: The number of ASCs and CGDs expected in the Vantage System 5. If the Vantage Hardware is not detected in the Device Manager, proceed to Step 3. 6. If the Vantage Hardware is detected, but the count of CGDs is incorrect, proceed to Step 5.	Vantage System Configuration	ASCs	CGDs	Vantage 64	1	2	Vantage 64LE	1	4	Vantage 128	1	4	Vantage 256	1	8
Vantage System Configuration	ASCs	CGDs															
Vantage 64	1	2															
Vantage 64LE	1	4															
Vantage 128	1	4															
Vantage 256	1	8															

Step#	Issue	Troubleshooting Procedure
3	<i>The Verasonics Hardware is NOT detected in the Device Manager, but "PCI Data Acquisition and Signal Processing?" is listed in the Device Manager.</i>	Install the correct device driver. - Follow software installation instructions from the Vantage System User Manual (located in the Documentation folder), to install WinDriver. - After installing the device driver, the user should see Verasonics Hardware in Device Manager. - If the issue is not resolved, proceed to Step 4.
4	<i>The Verasonics Hardware is NOT detected in the Device Manager, and an error message, "Hardware NOT found" is displayed.</i>	Inspect the PCIe cable for an incomplete or failed connection. - Check the PCIe slot and PCIe connector for any debris and clean it appropriately. - Plug in the PCIe adapter securely by pressing it into the PCIe connector slot until you hear a distinct clicking sound. - Note that there are two green LED lights on the PCIe adapter—a solid green LED light to the left indicates a successful express link with the Vantage hardware system and a solid/blinking light to the right indicates a successful express link with the host controller.  Two green LEDs indicating a successful PCIe connection <i>Figure 2: Two green LEDs on the backside of the Vantage Hardware System</i> - If only one of the LEDs lights up, it indicates that the PCIe adapter card is not well seated in the slot. Remove the PCIe adapter from the slot and reinstall it. - If either one of the LEDs is red, it indicates a power fault—ensure that nothing is touching the module inside the computer. If the issue is not fixed, either the PCIe cable, PCIe adapter, I/O panel, and/or the backplane in the system needs replacement. Contact Verasonics Support (support@verasonics.com) for more information.

Step#	Issue	Troubleshooting Procedure
5	<i>The Verasonics Hardware is detected in the Device Manager with an incorrect number of CGDs, and an error message, "The hardware can be detected but cannot be initialized or the system software is unable to communicate with it." is displayed.</i>	Reprogram Verasonics Hardware. 1. Perform a cold boot on the host controller and the Vantage hardware system. 2. Open MATLAB and navigate to the <i>Vantage Project</i> folder. 3. In the MATLAB Command Window, type <code>activate</code>, and press Enter. 4. Type the <code>reprogramHardware</code> command and press Enter. Pay attention to all the prompts and status updates displayed in the MATLAB Command Window. 5. Note that the <code>reprogramHardware</code> command forces revision checks and updates to all FPGA code files in the Vantage hardware system and this process may take several minutes to complete. Note: Do not interrupt the system while the update process is in progress. 6. Shut down the host controller to a full power-off state—a full power cycle is required for the FPGA file updates to take effect. 7. Power off the Vantage hardware system by pressing the Power Off button and wait for 10 seconds before turning it back on. 8. First power on the Vantage hardware system and then power on the host controller. Note: The Vantage hardware system should ALWAYS be powered on before powering on the host controller. 9. Open MATLAB and navigate to the <i>Vantage Project</i> folder. 10. In the MATLAB Command Window, type <code>activate</code>, and press Enter. 11. Type <code>VerasonicsVerificationTest</code>, and press ENTER. 12. If the tests pass, try running a script. Note: If the <code>reprogramHardware</code> command from the current software version does not fix the issue, update to the latest software version, and perform the above procedure again. If an error occurs after <code>reprogramHardware</code> Command is issued, proceed to Step 6.

Step#	Issue	Troubleshooting Procedure
6	<i>If an error message occurs during or after the reprogramHardware command is issued and an error message on ASC is received.</i>	Reprogram Verasonics Hardware in Recovery mode. - While the Vantage hardware system is on, toggle the system into Recovery mode. - Using any small, pointed tool (such as a paper clip), push the Toggle button. - Note that the LED turns red to indicate that the system is in Recovery mode.  Figure 3: Toggle button at the backside of Vantage Hardware System  Figure 4: Green LED indicates that the system is operating in normal mode  Figure 5: Red LED indicates that the system is operating in recovery mode - Power off the host controller and then power off the Vantage hardware system. - Power on the Vantage hardware system and then power on the host controller. Note: The Vantage hardware system should ALWAYS be powered on before powering on the host controller. - Open MATLAB and navigate to the <i>Vantage Project</i> folder. - In the MATLAB Command Window, type <code>activate</code>, and press Enter. - Type <code>reprogramHardware</code>, and press Enter. - During the process of reprogramming the hardware, you will be prompted to toggle the system back to Normal operating mode. The LED turns green indicating that the Vantage Hardware System is now operating in Normal mode. - Save any error messages.

Step#	Issue	Troubleshooting Procedure
		9. Shut down the host controller to a full power-off state—a full power cycle is required for the FPGA file updates to take effect. 10. Shut down the Vantage hardware system. 11. Power on the Vantage hardware system and then power on the host controller. 12. Open MATLAB and navigate to the <i>Vantage Project</i> folder. 13. In the MATLAB Command Window, type <code>activate</code> , and press Enter. 14. Type <code>VerasonicsVerificationTest</code> and press Enter. 15. If the tests PASS, try running a script. If you get an ACI or CGD error during or after running <code>VerasonicsVerificationTest</code> , proceed to Step 7 .
7	If Verasonics Hardware is detected in Device Manager, but you get an ACI or CGD error while doing the Verasonics Verification Test.	Check to see if there is a requirement for board replacement. - Check whether the green LED are blinking on all boards.  Green LED indicating a successful express link for the CGD on Board 1. <i>Figure 6: Green LEDs on the boards as seen from the backside of the Vantage Hardware System</i> - If any green LED is OFF, check the red LED about halfway down the backplane of the same board. - If the red LED is ON, contact Verasonics Support (support@verasonics.com) for board replacement. - If the red LED is not ON, but the issue persists, proceed to Step 9.
8	The ID of the transducer specified by	Ensure that the transducer is connected to the connector. Re-plug the transducer or restart the host controller.

Step#	Issue	Troubleshooting Procedure
	the script does not match the ID of the transducer at connector 1.	2. If the issue persists, plug different probes and type "ProbeConnectorStatus" in the MATLAB Command Window for each probe. If all of them do not work, send a log file to the Verasonics Support team. Proceed to Step 9.
9	Contact Verasonics Support with troubleshooting logs.	Send an email to the Verasonics Support team, support@verasonics.com. Attach the following test results to your email: ▪ Version X text file ▪ systemInfo ▪ VVT Verbose test results ▪ reprogramHardware X test results ▪ [failed, output] = hwdiag('microregistersdump') test results Refer to Section 2 for more details on collecting the above troubleshooting logs and test results.

2. Collect Troubleshooting Logs and Test Results

Run the following commands and attach the corresponding **six output files** to the email you send to the Verasonics support team (support@verasonics.com).

2.1 Version X

1. Open MATLAB and navigate to the *Vantage Project* folder.
2. In the MATLAB Command Window, type `activate`, and press ENTER.
3. Type `Version X`, and press ENTER.
4. Copy the output of the `Version X` command into a text file; attach this text file to your email.

2.2 systemInfo

1. Type `systemInfo` command, and press ENTER.
2. Note that the `systemInfo.txt` file is generated at the root of the Vantage project.
3. Attach the `systemInfo.txt` file to your email.

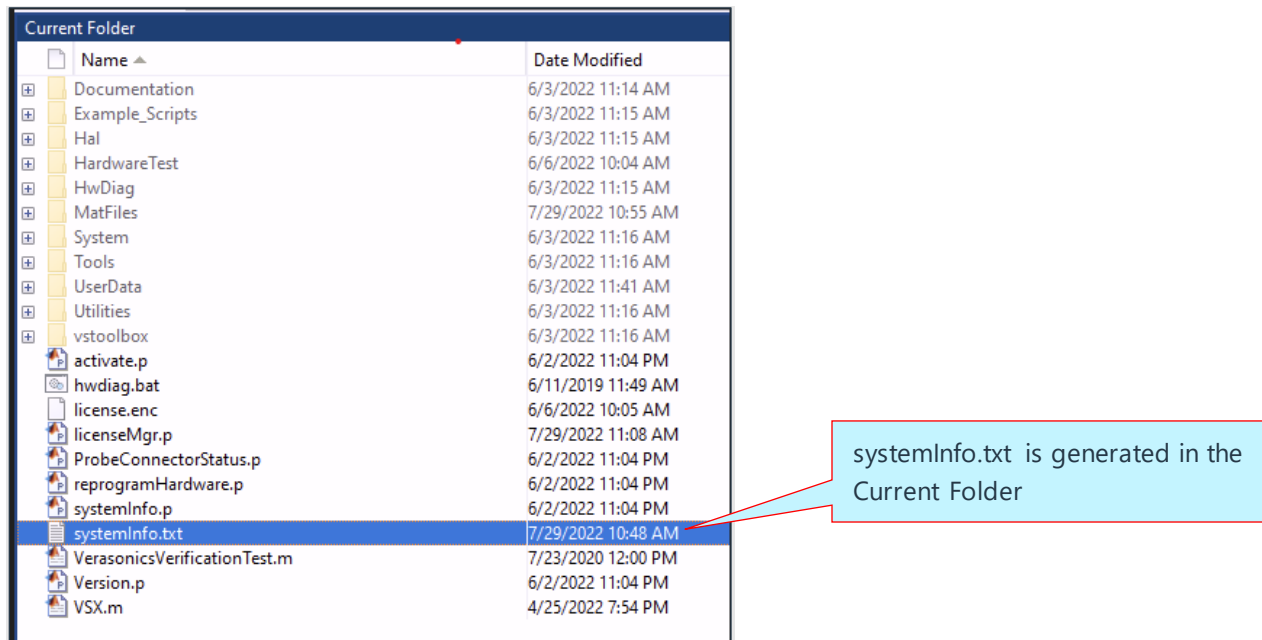
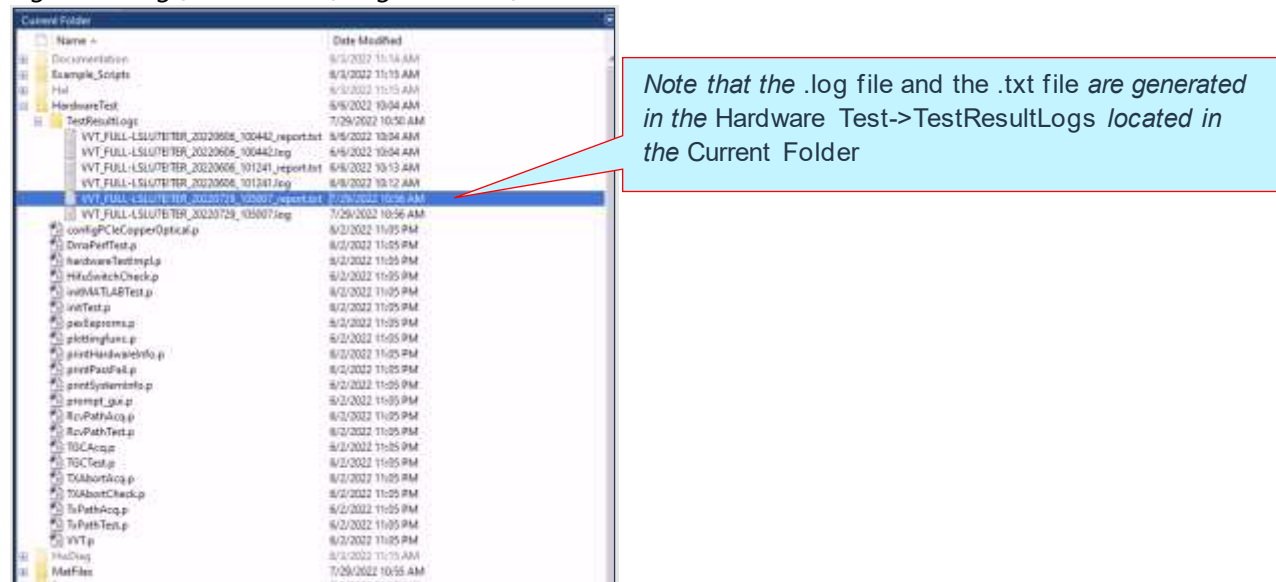


Figure 7: `systemInfo` file generated at the root of the Vantage directory

2.3 VVT Verbose test results

1. Disconnect any transducers.
2. Shut down the host controller to a full power-off state—a full power cycle is required for the FPGA file updates to take effect.
3. Shut down the Vantage hardware system.
4. Power ON the Vantage hardware system first and then power ON the host controller.
5. Open MATLAB and navigate to the *Vantage Project* folder.
6. In the MATLAB Command Window, type `activate`, and press ENTER.
7. Type `VerasonicsVerificationTest`, and press ENTER.
8. On the VVT prompt, check the box under “Test Options” that says, “Verbose Mode”, and then click the “Vantage System Test” button.
9. Note that the test results files are saved in the *TestResultLogs* folder located in the *HardwareTest* folder.

Figure 8: .log file and .txt file generated for the current run



- Copy and save the .log file and the .txt file (for the run of VVT in verbose mode); attach these files to your email.

2.4 reprogramHardware X

- Perform a cold boot on the host Controller and the Vantage hardware system.
- Open MATLAB and navigate to the *Vantage Project* folder.
- In the MATLAB Command Window, type `activate`, and press ENTER.
- Type `reprogramHardware X` command and press Enter. Pay attention to all the prompts and status updates displayed in the MATLAB Command Window.
- Note that the `reprogramHardware X` command forces revision checks and updates to all FPGA code files in the Vantage hardware system and this process may take several minutes to complete.
Note: Do not interrupt the system while the update process is in progress.
- Shut down the host controller to a full power-off state—a full power cycle is required for the FPGA file updates to take effect.
- Shut down the Vantage hardware system.
- Power On the Vantage hardware system first and then power on the host controller.

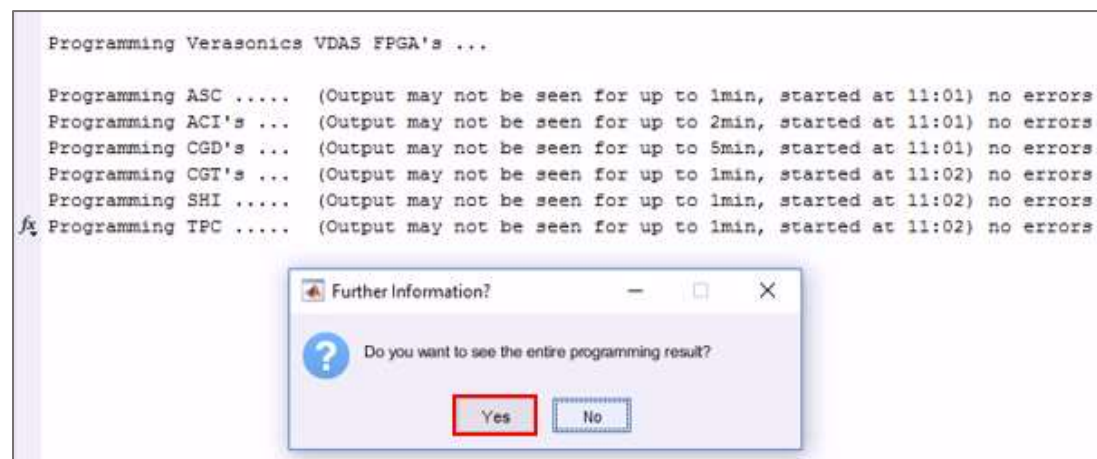


Figure 9: View detailed results by selecting 'Yes'

9. In the dialog box that pops up, select 'Yes' to see the entire programming result.
10. Copy the test result to a text file; attach this text file to your email.

2.5 hwdiag ('microregistersdump')

1. In the MATLAB Command Window, type activate, and press ENTER.
2. Type `[failed, output] = hwdiag('microregistersdump')`, and press ENTER.
3. Copy the test results to a text file; attach this text file to your email.

2.6 Advanced Debugging

Occasionally, the user may be asked by Verasonics support to acquire a detailed log file to support the investigation of problems in the system software. The system logging is enabled with the following command:

```
vsv.common.util.Logger.start()
```

The logger is stopped with the following command:

```
vsv.common.util.Logger.stop()
```

The log file is created at the root of the software installation. The log file is named using the pattern `VSV_YYYYMMDD_HHMMSS.log`, where `YYYYMMDD_HHMMSS` is the timestamp of the logger start time.

Typically, the following sequence of commands will be issued before sending the log file to Verasonics support.

```
vsv.common.util.Logger.start()
```

```
VSX
```

<Terminate script execution>

```
vsv.common.util.Logger.stop()
```

The log file grows large very quickly so the script should typically only be executed for the minimum amount of time to reproduce the behavior corresponding to the trouble report.

3. Troubleshoot Software Issues

Error Messages/Issues	Troubleshooting Procedures
"Your current directory is not set to the root of a Vantage Project software directory"	1. Access the Vantage Software installation files using your customer account using the file path - <i>Customer Download\Release 4.7.6\Vantage-4.7.6-2206101100</i>. Note: The main directory containing all Vantage software installation files (example: Vantage-4.7.6-2206101100), is referred to as the <i>Vantage project</i> folder across all Vantage system documents. 2. Select the <i>Vantage project</i> folder as the current MATLAB working directory.
Errors seen due to incorrect software installation procedure.	1. Follow software installation instructions from the Vantage System User Manual. Refer to the Vantage Release Notes for the current Vantage Software release while installing any software to ensure that the correct version of MATLAB and OS are installed on the host controller. 2. Open MATLAB and navigate to the <i>Vantage Project</i> folder. 3. In the MATLAB Command Window, type <code>activate</code>, and press Enter. Only after executing the <code>activate</code> command can you run any other commands.
Errors seen when the script (you are trying to run) cannot be not found	1. Navigate to the folder where the setup script is located. 2. Run the script and select "Add to Path". This allows the user to run a script that is not shown in the MATLAB search path.
Errors seen due to MATLAB version incompatibility with the current Vantage Software Release	1. Find the list of supported versions of MATLAB for the current Vantage software release version in the Vantage Release Notes. 2. Confirm whether the MATLAB version that is being used is supported by the current Vantage software release. 3. Update the MATLAB version to one of the supported versions.
MATLAB crashes, even before VSX loads a .mat file	Such errors are typically caused due to a conflict between the Vantage software and third-party MATLAB toolboxes, such as the MATLAB Compiler toolbox or functions in your MATLAB environment. These toolboxes may be corrupting your startup.m file. Resolve such issues by performing the following steps: 1. Save your original startup.m file as a different file name. 2. Disconnect all hardware. 3. Open MATLAB, without loading the startup.m function. 4. Change the current folder to the <i>Vantage Project</i> folder. 5. Type <code>activate</code>, and press Enter. 6. Ensure that only the <i>Vantage Project</i> folder and the default MATLAB Toolboxes are in the correct path. 7. Run VSX and load any .mat file. If problems still persist, install MATLAB with signal toolbox only. Other toolboxes are not required.

Error Messages/Issues	Troubleshooting Procedures
<i>Errors seen due to licensing issues</i>	If the Vantage system is not operable because of a missing, or incorrect license file, refer to the Software Licensing section in the Vantage System User Manual. Attach the systemInfo.txt file to an email message. In the email message, describe the situation briefly and send the email to licensing-request@verasonics.com.
<i>Errors seen when a third-party host controller is used by multiple users</i>	If you are using a third-party host controller (a host controller that is not supplied by Verasonics), ensure that it has a single master administrator. Note that adding additional users to the third-party host controller can cause problems with the Vantage software.
<i>Errors/crashes seen when unrelated applications are installed on the host controller</i>	The host Controller should be a dedicated resource of the Vantage hardware system—installing any unrelated applications on it can cause the system to crash.
<i>Errors seen when an incorrect 'timeToNextAcq' value is set</i>	The 'timeToNextAcq' value is the time in microseconds from the start of the transmit period in the event that it is specified to the start of the transmit period. The minimum time for a transmit/receive acquisition is around 10usec (the actual minimum is 128 A/D samples or 10usec, whichever is greater), so if you set the 'timeToNextAcq' value smaller than something in this range and the timing criteria is not met, an error is produced.
<i>Warnings indicating missed 'timeToNextAcq'</i>	Usually, the missed 'timeToNextAcq' warnings are due to changing the high voltage. The amount of time missed is consistent with the time the high voltage supply takes to slew to a new level. If the high voltage change occurs while the hardware sequencer is paused (which is likely the case at the 'sync' point) it may take a transmit event to initiate the change in high voltage. In addition, a Direct Memory Access (DMA) data transfer can only launch when the previous DMA has been completed; if the previous DMA is still in progress, the hardware sequencer pauses until it is complete.
<i>Warnings or errors due to different limits while using Profile 5</i>	When using Profile 5 for extended transmit applications, the following warning or error messages related to the hardware limits may appear: ▪ Transmit flux limit ▪ Peak-transmit output current limit ▪ Gate-driver supply current limit ▪ Gate driver device estimated temperature rise ▪ TX FET temperature rise ▪ Push Capacitor Peak current ▪ Estimated average HV transmit supply

Error Messages/Issues	Troubleshooting Procedures
	Refer to <code>computeTWWaveform.m</code> and <code>TXEventCheck.m</code> to find all the active transmit events in the event sequence as defined by the user setup script, estimate the operating state currents, dissipation levels, etc., and compare them to the limits that have been defined to ensure safe, reliable operation of the system. In order to avoid reaching limits, the possible options to choose from are as follows: ▪ Lower the transmit frequency ▪ Reduce the active elements by changing TX.Apod setting ▪ Reduce the duty cycle ▪ Change the transmit pulse length
<i>Installation failure for WinDriver</i>	The user should be able to reset the NVIDIA files by navigating to <code>/usr/src/nvidia<versionnumber></code> and running the following commands. <pre>sudo make clean sudo make</pre> This ensures that all the latest needed drivers are built. The user can now retry the WinDriver installer.
<i>Any other errors</i>	Use the Event Analysis Tool to find possible script errors before asking for support. Refer to the Application Note – Event Analysis Tool for more information. If you can't identify where the bug is, please email your script and output of <code>systemInfo</code> to Verasonics Support at support@verasonics.com. Free support is limited to 8 hours per system per SLA per year.